
Team 403 – Design Report

CIV102: Structures and Materials – An Introduction to Engineering Design

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Introduction

Overview

This report discusses major iterations of Group 403's final proposed bridge and details the calculations and decisions made before settling on the final design. This includes all 5 iterations and their associated reason for changing any specific parts, the specific changes made, mathematical justifications for changes, the results (specifically stresses and factors and safety) and new mode of failure and associated factor of safety. Figures and/or diagrams will be included as either evidence of calculations or visualization of the new design after every iteration.

Background: Design 0

Before the iteration started, an analysis both by hand and the code was run, and summarized below are the key takeaways:

```
Load Case 1
FOS for tension 4.3621
FOS for compression: 1.0375
FOS for flexural buckling: 0.59735
Case 1 buckling
FOS for shear: 2.667
FOS for glue: 9.3863
FOS for shear buckling: 3.391

Load Case 2
FOS for tension 3.8787
FOS for compression: 0.92254
FOS for flexural buckling: 0.53116
Case 1 buckling
FOS for shear: 2.2568
FOS for glue: 7.9427
FOS for shear buckling: 2.8695
```

Figure 1: All the Factors of Safety (FOS) for Design 0 under Load Case 1 and 2 (Base case). Note especially that $FOS_{flexural\ buckling} < 1$ and $FOS_{compression}$ is close to 1 for Load Case 1.

The Failure Case for Design 0 is the buckling of the side webs, with an FOS of 0.597. In theory, Design 0 would fail right away in a 400 N Load Case 1. Besides, the FOS for compression is very close to 1. This indicates that our first iteration step must address the issue of flexural buckling of the flange between the webs and potentially also increase the FOS for compression.

Similarly, the FOS for glue is relatively high at 9.39. This indicates that the FOS can be lowered for either material conservation or other structural purposes.

Final Bridge and Key Design Decisions

The final bridge contains a constant cross-sectional area throughout the bridge, Refer to **Figure 2**, this has a top flange of length 100 mm and height 3.81 mm. This also has a glue tab length of 10 mm and a bottom flange length of 75 mm. All other parts have a thickness of 1.27 mm.

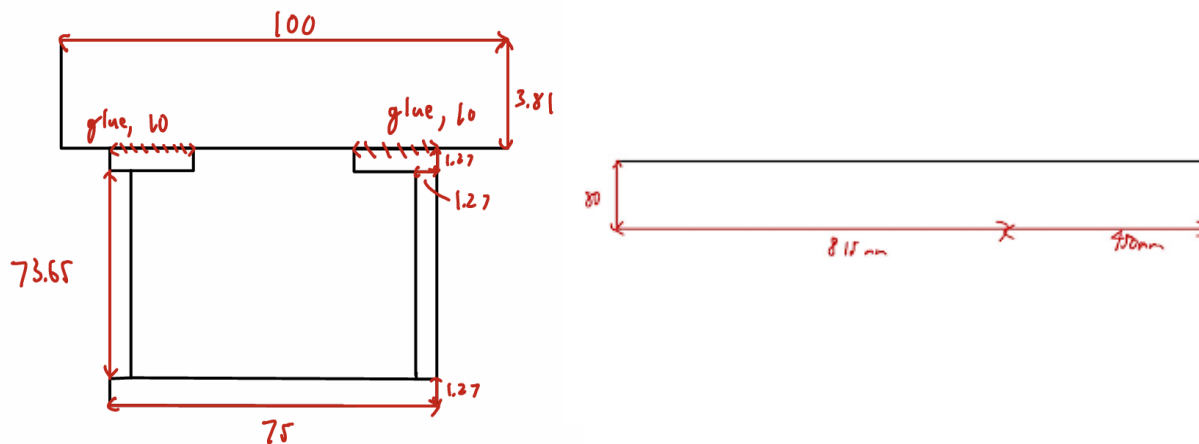


Figure 2:

(Left): The cross-sectional view of the bridge. Note specifically it follows a similar geometry to Design 0 with edited numerical values specifically chosen.

(Right): The side view of the bridge and how it is divided. The total length is 1265mm with height of 80mm.

It also has a splice connection at 450 mm, with a split of 450 mm and 815 mm portions, this accounts to a total length of 1265 mm. This is found out to be a more efficient way to cut the matboard and to utilize the materials. It also has a height of 80 mm to follow the “multiple of 20 mm” constraint on height listed on the original assignment. This bridge has a predicted failure load of 930 N under Load Case 2 when the lightest freight car weighs 270 N.

Below is the list of Key Design Decisions that are determined to reach this final design:

1. Our team decided that the bridge should at least (determined via calculations) pass Load Case 1 and the base case of Load Case 2. In other words, all the Factors of Safety for both cases for our final design shall be at least > 1 .
2. Our team also decided that we should not consume all of the materials for the build as one of the major evaluation criteria of the bridge is the strength to weight ratio. The size of the matboard also restraints the material we use and the cut we decide to implement on the matboard.
3. The final decision made concerns about the complexity of the structure (and especially the cross-sectional area). This decision is made to address the quality mark of our bridge. The team has decided to limit the geometry to consist of only rectangles to maximize the building quality. This also considers Section 2.7 of the original Handout, where no calculations for splice connections are present for CIV102.

The cross-sectional area is designed to meet all these Design Decisions. The thick top flange is implemented to maximize the FOS. Heavy glue usage is present to counteract with failure modes not addressed by calculations of CIV102. It was later determined that a constant cross-sectional area is more desirable in terms of lowering complexity (See Iteration 5). Besides, a design with constant cross-sectional area allows the cut to be more easily set up, saving more materials and lowering the complexity.

Major Iterations

All FOSs shown below are based on Load Case 1 and the base case of Load Case 2. For analysis purposes, the following terms and variables are defined:

- Case 1 buckling refers to the plate buckling of the middle of the top flanges
 - Case 2 buckling refers to the plate buckling of the free sides of the top flanges
 - Case 3 buckling refers to the plate buckling of the stress profiles of the webs.
-
- I is the Second Moment of Inertia
 - y_{bar} is the centroidal distance from the bottom of the cross-section to the centroid.
 - Q is the First Moment of Area
 - t is the thickness of the flange when considering plate buckling
 - b is the base width depending on Case 1,2,3 or shear buckling when considering plate buckling

Design Iteration 1

Reason:

Upon analyzing decision 0 under a moving train in Load Case 1, the lowest FOS is determined to be $0.597 < 1$ for **flexural buckling**, the failure mode being **Case 1 Buckling**.

Change(s) made:

To increase the buckling stress due to **Case 1 Buckling**, refer to **Figure 3**, our team **reduced the base length of the cross section to 60 mm**.

This iteration considers **everything else to remain constant**, including the height of the bridge, top flange length, constant cross-section. This will most likely decrease material required so material is also not considered. Diaphragms will also be spaced 400 mm apart as in Design 0.

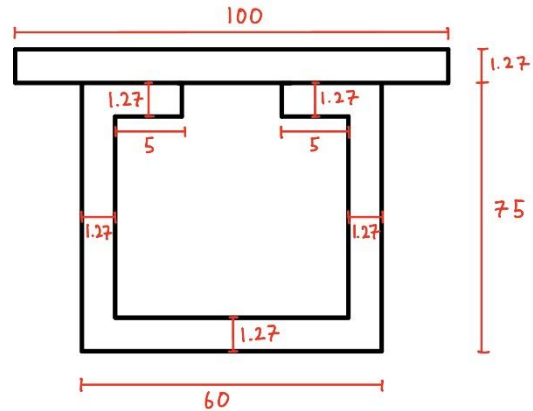


Figure 3: The cross section for Iteration 1. This is essentially the same as Design 0 with decreased bottom flange width of 60 mm.

Justification:

The plate buckling equation is proportional to the expression $(t/b)^2$, where t is the thickness of the top flange and b is the base length of the flange in between webs in this case, i.e. reducing the base length leads to increasing plate buckling stress.

Results:

By reducing b by $\frac{3}{4}$ the flexural stress was increased by a factor of $(4/3)^2 = 16/9$ times. The major changes for factors of safety (FOS) are the following:

The lowest FOS is still the FOS for **flexural buckling** but is still < 1 . This factor of safety is now governed by **Case 2 Buckling**. This is because reducing the base length increases the length of the free edges subjected to flexural buckling. Thus, decreasing the allowable stress and strength of these free edges.

```
Load Case 1
FOS for tension 3.6661
FOS for compression: 0.9998
FOS for flexural buckling: 0.97952
Case 2 buckling
FOS for shear: 2.6217
FOS for glue: 9.0587
FOS for shear buckling: 3.3334

Load Case 2
FOS for tension 3.2599
FOS for compression: 0.88902
FOS for flexural buckling: 0.87099
Case 2 buckling
FOS for shear: 2.2185
FOS for glue: 7.6655
```

Figure 4: All FOS for Iteration 1. It indicates Iteration 1 fails in Load Case 1 by flexural buckling Case 2.

Since I decreased, and y_{bar} shifted upwards. Two major effects are induced:

1. Increased the bottom tension and decreased the FOS from a 4.36 to a 3.67, this is desirable as 4.36 is slightly too high.
2. FOS for compression decreased slightly from 1.07 to 0.999, this is not desirable as a FOS for compressive buckling is now < 1 and bridge can now fail by compressive buckling as well.

All other FOS values are high enough to be deemed safe and have minimal changes. In conclusion, it is clear that the changes solely made to edit one parameter of the geometry of Design 0 will not work as desired. Multiple factors such as I , and length of free top flanges need to be balanced at the same time. **Future Iterations must change multiple values at once.**

New Failure Case:

At Load Case 1, the bridge would fail by either **buckling of the free flanges** or **compressive buckling**. Train weight is 400 N.

Design Iteration 2

Reason:

After Iteration 1, the FOS for flexural buckling and compressive buckling was lower than the minimum FOS of 1. Iteration 2 should increase this to a minimum of 1. Additionally, the FOS for glue was **unnecessarily high**, Iteration 2 shall reduce that.

Change(s) made:

Refer to **Figure 5**, the thickness of the top flange was doubled by adding another layer of matboard making the thickness 2.54 mm, and the width of the bottom flange was increased to 75 mm.

Decreased the glue tab width (and consequently length of glue applied) to 3 mm in order to get rid of unnecessary material use and safety.

It is also hypothesized that the compressive stress can be increased to ensure a FOS for compressive force of at least 1 by increasing the total height to 123.81 mm. (Referring to the multiple of 20 height constraint).

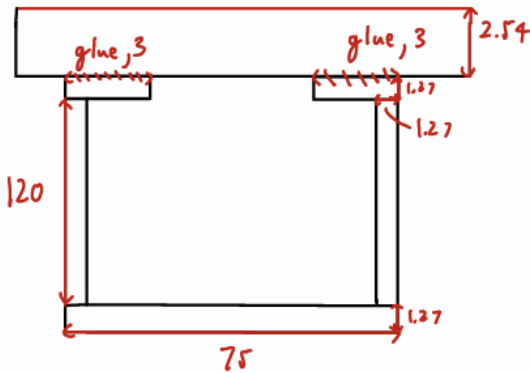


Figure 5: The cross section for Iteration 2. This design increases the web height to 120 mm, top flange height to 2.54 mm and bottom flange width of 75 mm. The glue tab width and applied distance is decreased to 3 mm.

Justification:

The equation to calculate **Case 2 Buckling** includes the factor $(t/b)^2$. By doubling the thickness, the Case 2 Buckling stress increases by a factor of 4. The bottom flange was also increased to 75 mm so that the provided dimensions of the distance between the wheels are the same as the width of the bottom flange.

As for compressive stress, by increasing the height of the cross section, I increases. Similarly, y_{bar} should be situated at a higher position, reducing the compressive stress on the stress profile. Compressive stress is also inversely proportional to I . This lowers the compressive stress experienced by the bridge.

Results:

By increasing the thickness of the top flange, height of the bridge and decreasing the glue tab's thickness, refer to **Figure 6**, the following major changes occurred for FOSs:

Both flexural stress' factors of safety significantly increased. This is due to the increased I value. Maximum moment is kept unchanged, y_{bar} is shifted upwards, inducing **more tensile stress** and **lowering compressive stress** in the flexural profile. This significantly improves the bridge's performance as the FOS for both tension and compression are now > 1 .

By shifting y_{bar} upwards and doubling the top flange's thickness (t), the plate buckling stress for Case 2 is

```

Load Case 1
FOS for tension 8.5212
FOS for compression: 2.765
FOS for flexural buckling: 7.2452
Case 1 buckling
FOS for shear: 4.1528
FOS for glue: 6.0395
FOS for shear buckling: 2.1718

Load Case 2
FOS for tension 7.577
FOS for compression: 2.4586
FOS for flexural buckling: 6.4424
Case 1 buckling
FOS for shear: 3.5141
FOS for glue: 5.1107
FOS for shear buckling: 1.8378

```

Figure 6: All FOSs for Iteration 2. It indicates Iteration 2 does not fail by Load Case 1 and Load Case 2. Note the lowest FOS is shear buckling.

significantly increased. The governing factor for FOS for flexural buckling is now **Case 1 buckling**, at a significantly higher 7.24.

The FOS for shear increased slightly due to the increased I value leading to **lower shear stress**. The FOS for glue decreases slightly as the base length (b) where the glue is applied decreases, this allows less glue usage for the same desired outcome.

Finally, the FOS for shear buckling decreases as both values of centroidal shear stress and stress of shear buckling changes. This decrease arises as the **centroidal shear stress increased more than the shear buckling stress** did.

In conclusion, this iteration successfully improved metrics in terms of flexural stresses and the shear stress of glue. It should be noted that material usage is significantly higher in this design (excluding glue) ~ 1.6 times of Iteration 1.

The dynamics between shear buckling and shear stress should also be investigated for Iteration 2.

New Failure Case:

The bridge should now fail in Load Case 2 by **shear buckling** when the lightest freight car has a weight of 240 N and the total train should be **828 N**.

Design Iteration 3

Reason:

Since the FOS for compression was the smallest FOS. After Iteration 2, our team hypothesized that using a non-constant cross-sectional area would maximize the stress capacity related to the SFE and BME. This is because the maximum bending moments occur in the middle of the track, placing heavy emphasis on minimizing materials.

Change(s) made:

Refer to **Figure 7**, our team split the cross-sectional area into 2 parts, including a more “material heavy area” in 401-800 mm and a “material light area” in 0-400 and 801-1200 mm. Both areas draw heavy reference from Design 0. A diaphragm length of 200 mm is also implemented to solidify the structure.

Justification:

Since the Maximum bending moment always occurs close to the middle of the beam. Refer to **Figure 7**, the material heavy area would **provide a much greater I value**, decreasing the experienced stress in the middle of the bridge.

Conversely, the lesser material edges' area would have a lower I value. This allows the **less critical areas to have a lower capacity which saves materials and still provide the necessary strength**.

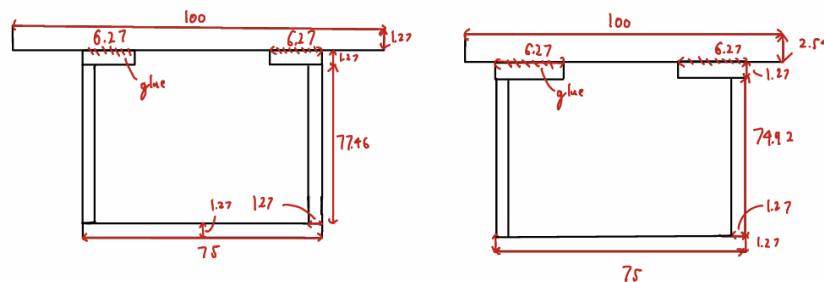


Figure 7: the cross sections for Iteration 3. This adapts the non-constant cross-sectional area design. The left area is the “material light area”, which has a similar area with Design 0 (Slightly increased web height). This addresses the issue of unnecessary strength by the sides. The right area is the “material heavy area”, which has a similar area to Iteration 2 (Reverted glue tab width and web height) This addresses the issue of high strength demand in the middle.

Results:

By splitting the cross-sectional areas, refer to **Figure 8**, the FOSs changes (According to Load Case 2) by the following:

The FOS for flexural stresses is lowered significantly. The tensile FOS is at a satisfactory 3.70 and the compressive FOS is at a slightly low 1.48. Since Iteration 2 took a lot of material, the I value is naturally high. As the cut in material is made, the I value drops significantly, **increasing both the tensile and compressive stress the beam experiences.**

Load Case 1
 FOS for tension 4.1587
 FOS for compression: 1.6688
 FOS for flexural buckling: 1.6349
 Case 2 buckling
 FOS for shear: 2.3406
 FOS for glue: 7.6156
 FOS for shear buckling: 3.0133

Load Case 2
 FOS for tension 3.6979
 FOS for compression: 1.4839
 FOS for flexural buckling: 1.4538
 Case 2 buckling
 FOS for shear: 1.9807
 FOS for glue: 6.4443
 FOS for shear buckling: 2.5499

Figure 8: All FOSs for Iteration 3, it indicates Iteration 3 does not fail in Load Case 1 and Load Case 2. Note the lowest FOSs are compression and flexural buckling by Case 2.

The FOS for flexural buckling is also significantly lowered to 1.45. Now governed by **Case 2 Buckling**. The decrease in the thickness (t) of the top flange for the edges of the bridge **lowers the Case 2 buckling Stress**.

Similarly, FOS for shear decreased to 1.98 as change in the centroidal shear stress is **proportional to Q** and **inversely proportional to I** , This indicates Q decreased lesser than I did, increasing the shear stress the cross section is subjected to.

Conversely, the FOS for glue **increases** to a 6.44 as the glue length is reverted to Design 0's. This is still unnecessarily high and **not a concern for failing cases**.

The FOS for shear buckling increased as the **shear buckling stress increased** since the **spacing between diaphragms decreased**. This increases it to 2.54 for which is a desirable number as it is not too high or too low.

In conclusion, the FOSs drop as **less material is used**, the FOS for shear buckling is desirable but the FOS for glue tab is not desirable. A proper justification needs to be provided or the **FOS for glue needs to be reduced for conserving material**.

The next iteration shall try to increase the FOSs in general within a reasonable number of materials.

New Failure Case:

The bridge should now fail in Load Case 2 when the lightest freight car has a weight of 135 N (lowest possible) and the total train should be **466 N**. Failure by **Case 1 Buckling**.

Design Iteration 4

Reason:

After Iteration 3, it was clear that a varying cross-sectional area allows better material allocation for different parts of the bridge. However, the cross-section must be carefully chosen to have a suitable FOS. As a result, Iteration 4 aims to increase every FOS value.

Change(s) made:

Iteration 4 attains the varying cross-sectional area and attempts to strengthen it by **using more materials**. Both areas still have the same total height of 80 mm.

Refer to **Figure 9**, for the “lesser material area”, the thickness of the top flange is increased to 2.54 mm; For the “material heavy area” the thickness of the top flange is now increased to 3.81 mm.

Both areas have doubled web thickness of 2.54 mm. The rest of the properties remain the same from Iteration 3.

The glue tab shall retain the same thickness as above as a “safety measurement” for increasing every metric.

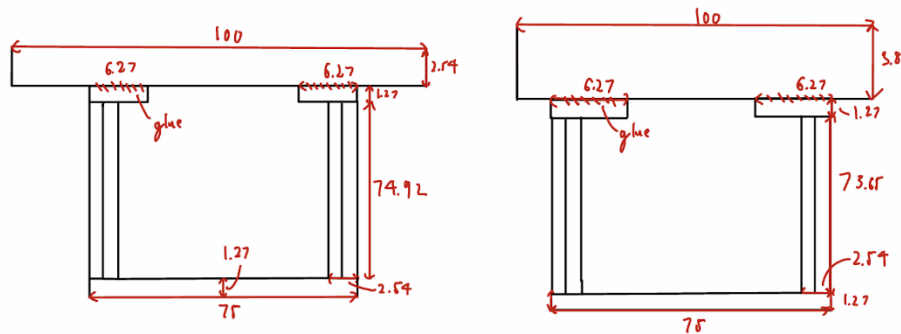


Figure 9: The cross sections for Iteration 4. This adapts the non-constant cross-sectional area design. The left area is the “material light area”, now with increased top flange height and double webs. This increases the strength. The right area is the “material heavy area”, now with increased top flange height and double webs, further increasing the strength in the middle.

Justification:

By increasing the thickness of the top flange and the webs, I is increased, it also **shifts y_{bar} upwards**. This also increases the thickness values (t) for plate buckling equations.

As a result of increases in these values, it is expected that all the stresses experienced are decreased and the maximum stress capacity is increased for all parts.

Results:

As a result of Iteration 4, all FOSs increased. (All analysis below is done in Loading Case 2)

The increase in the FOS of flexural stresses is due to the **increase in I and changing y_{bar} value**. This decreases the flexural stresses experienced by the bridge as a whole. FOS for tension and compression is now 5.59 and 2.11 respectively.

The increase in the FOS for shear force to 4.46 is due to the increase in I and Q value. The shear stress is directly proportional to Q and inversely proportional to I . Since **effects induced by increasing I were greater**, this **lowers the shear force experienced by the cross-sectional area** significantly. This also applies to the increase in FOS for glue to 7.23.

The increase in FOS of flexural buckling to 5.52 is **directly related to the increased thickness (t)** of the webs as Case 1 buckling stress capacities is **proportional to the flange thickness**; Case 2 and 3 buckling stress capacities are directly **proportional to the web thickness**.

Finally, the immense increase of FOS for shear buckling to 23.6 is related to the **increased web thickness**. This increases the shear buckling capacity of the cross section.

A critical result that is not calculation based identified would be that this design is **complicated to build**. As the board needs to be cut at a lot of positions and **multiple splice connections** need to be present.

In conclusion, all FOSs are either good or too high. This build is also **material heavy and hard to execute**.

New Failure Case:

The bridge should now fail in Load Case 2 when the lightest freight car has a weight of 275 N and the total train should be **950 N**. Failure by **compressive buckling**.

Design Iteration 5

Reason:

The team decided that a varying cross-sectional area is not desirable for the following 2 reasons:

Load Case 1	
FOS for tension	6.2948
FOS for compression	2.3714
FOS for flexural buckling	6.2138
Case 1 buckling	
FOS for shear	5.267
FOS for glue	8.5445
FOS for shear buckling	27.899
Load Case 2	
FOS for tension	5.5974
FOS for compression	2.1086
FOS for flexural buckling	5.5253
Case 1 buckling	
FOS for shear	4.4569
FOS for glue	7.2304
FOS for shear buckling	23.6083

Figure 10: All FOSs for Iteration 4, it indicates Iteration 4 does not fail in Load Case 1 and Load Case 2. Note the lowest FOS in Load Case 2 is compression and shear buckling has an unreasonably high FOS.

1. The **complexity of the build** and planned cuts required on the Matboard might **affect the quality** of the build.
2. No equations in CIV102 are present for calculating **splice connections**. The different cross-sectional area is hypothesized to lower the strength of that connection.

A **suitable FOS with proper material balancing** is required which is hard to achieve with the given materials.

Change(s) made:

As a result of the above judgment, a constant cross-sectional area is applied once again. Refer to **Figure 11**, consider this cross-sectional area a direct **improvement on Iteration 2** that **draws inspiration from Iteration 4**.

This increases the glue tab thickness to 10mm and the top flange height to 3.81mm. It also changes the diaphragm spacing to 150mm. Any other properties are kept constant. This also decreases the total height back to 80mm.

Justification:

As seen in **Iteration 4**, increasing the top flange height increases flexural FOSs due to **increased I value** and **y_{bar} shifting upwards**.

The glue tab width increase is due to a “**safety consideration**”. This is done to account for the chances that the glue does not dry by the testing date and other factors labelled in Section 2.7 of the project assignment (Splice connections, rigid cross-sections, support reactions.)

Although the decrease in total height decreases, I value significantly, this must be done in consideration to the **material constraints**.

Results:

Since this is a direct improvement on Iteration 2, all FOSs will be compared to Iteration 2. All of the analysis below is based on Load Case 2.

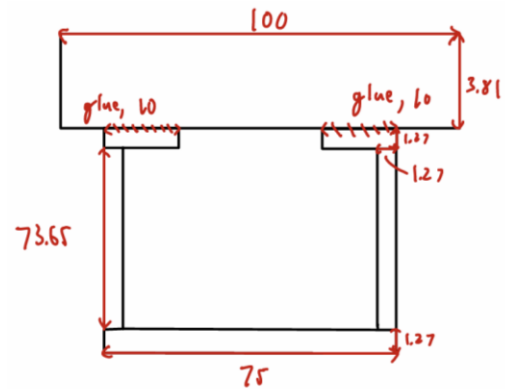


Figure 11: The cross section for Iteration 5. This re-adapts the constant cross-sectional area design. This is like Iteration 2 and draws inspiration from Iteration 4. This includes a tripled top flange height and immense increase in glue tab width. The total height is 80 mm.

FOS for tension decreased by a lot but is still at a high 4.31. This is due to **changes in I and y_{bar}** . However, the number is still high enough to be **deemed safe**.

FOS for compression decreased slightly from 2.46 to 2.06. This is the governing FOS for the bridge as it is the lowest. The change is due to the **decrease associated with I** .

FOS for flexural buckling is still governed by **Case 1 Buckling**. The increase in FOS to 12.153 is due to the **increased thickness (t)** of the top flange. This increases the buckling stress capacity of the cross section. This number is **significantly large** enough to be ignored.

FOS for shear decreases slightly to 2.30. This is due to the **decrease in I and Q , increasing the shear stress** experienced by the cross section. FOS for glue increases significantly as the **immense change in glue width** induces greater effects compared to the decrease in I and Q . The FOS for glue is now 10.1.

FOS for shear buckling increased significantly from a 1.84 to a 3.14. This is due to the combined effect of **closer diaphragms** and slight **decrease in the centroidal shear stress** mentioned above.

In conclusion, the bridge has an **overall suitable FOS** (statement stands for FOS of glue as it is done to account for non-calculated properties). This design is also viable to be drawn on the Matboard for cutting purposes and is suitable material wise.

New Failure Case:

The bridge should now fail in Load Case 2 when the lightest freight car has a weight of 270N and the total train should be **930 N**. Failure by **compressive buckling**.

Load Case 1
 FOS for tension 4.8431
 FOS for compression: 2.3182
 FOS for flexural buckling: 13.6674
 Case 1 buckling
 FOS for shear: 2.7214
 FOS for glue: 11.9087
 FOS for shear buckling: 3.7092

Load Case 2
 FOS for tension 4.3065
 FOS for compression: 2.0613
 FOS for flexural buckling: 12.153
 Case 1 buckling
 FOS for shear: 2.3029
 FOS for glue: 10.0772
 FOS for shear buckling: 3.1387

Figure 12: All FOSs for Iteration 5, it indicates Iteration 5 does not fail in Load Case 1 and Load Case 2. Note the lowest FOS in Load Case 2 is compression, at 2.06. The FOS for glue is high to account for failure modes besides calculations.

References

1. CIV Course Team, “CIV102 Matboard Bridge Design Project: Fall 2024”, downloaded from Quercus, 2024
2. MathWorks, “MATLAB - MathWorks,” *Mathworks.com*, 2019.
<https://www.mathworks.com/products/matlab.html>